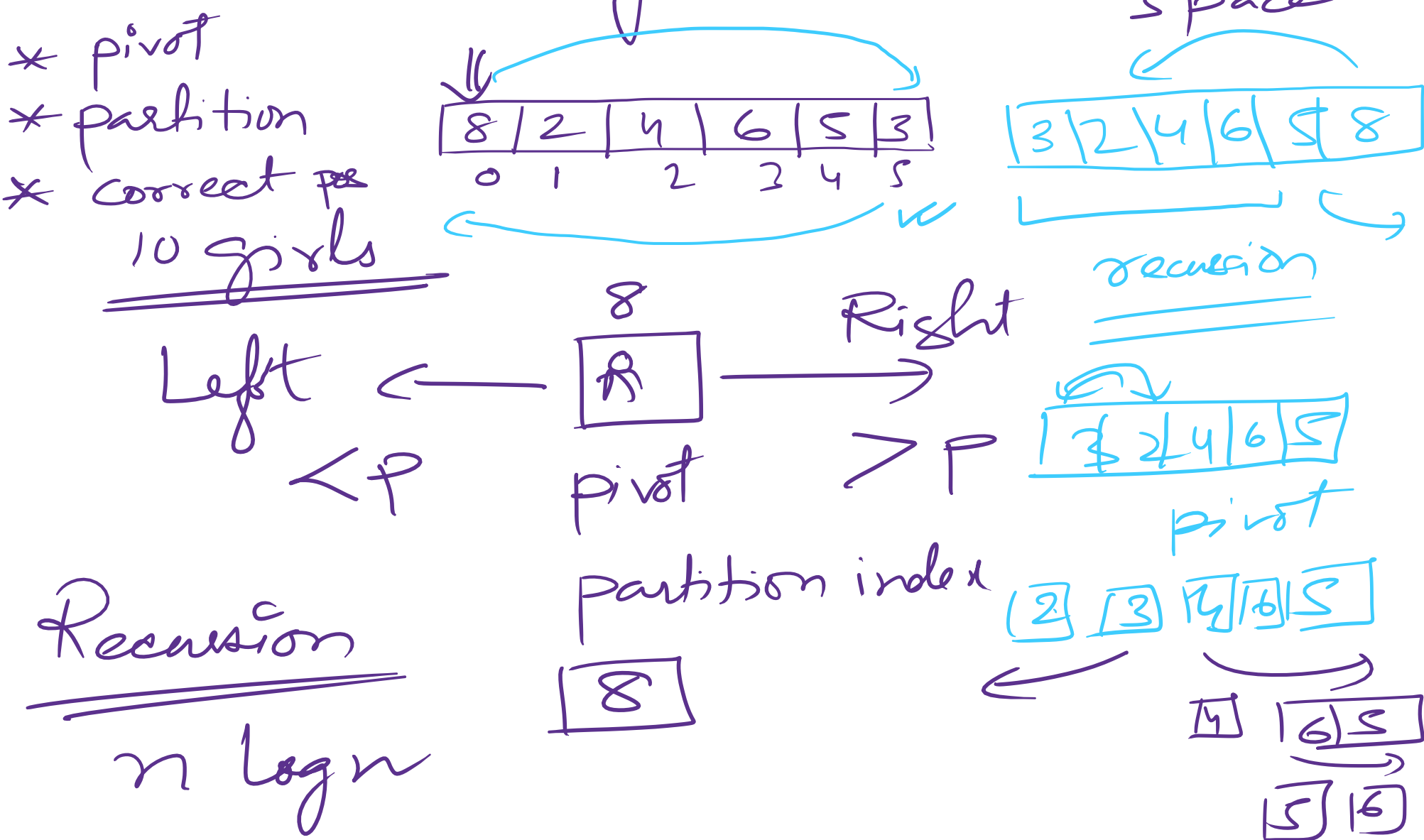


Trees, Binary Trees, Binary Search Trees
Traversals $\rightarrow$ Level Order, In-Pre-Post
Problems.

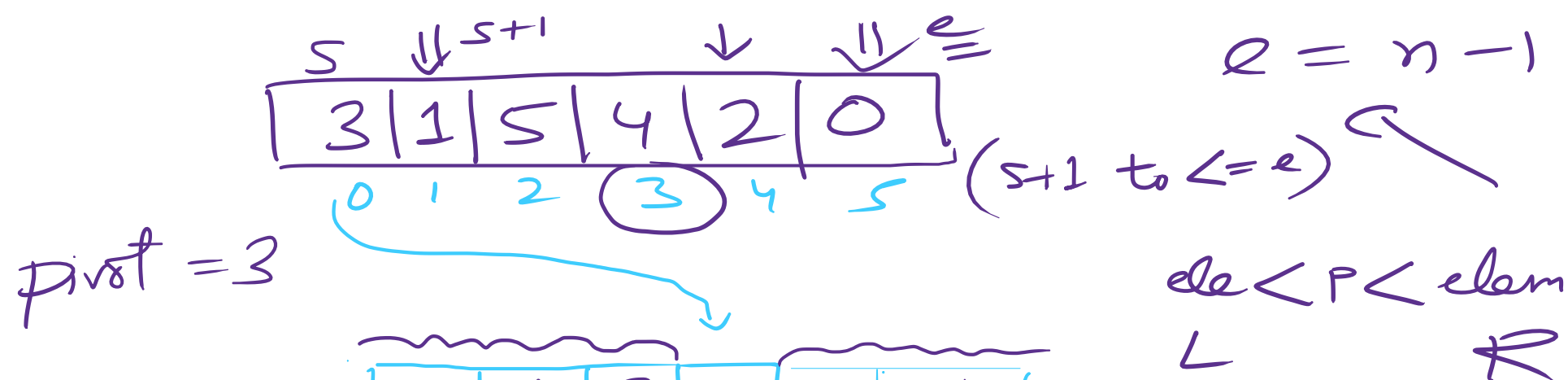
- * Quick Sort
- * Wave Sort
- * Standard Template Libraries

The Quick Sort Algorithm $\rightarrow$ No extra space

- * pivot
 - * partition
 - * correct ~~pe~~
- 10 girls



Recursion
 $n \log n$



while ()

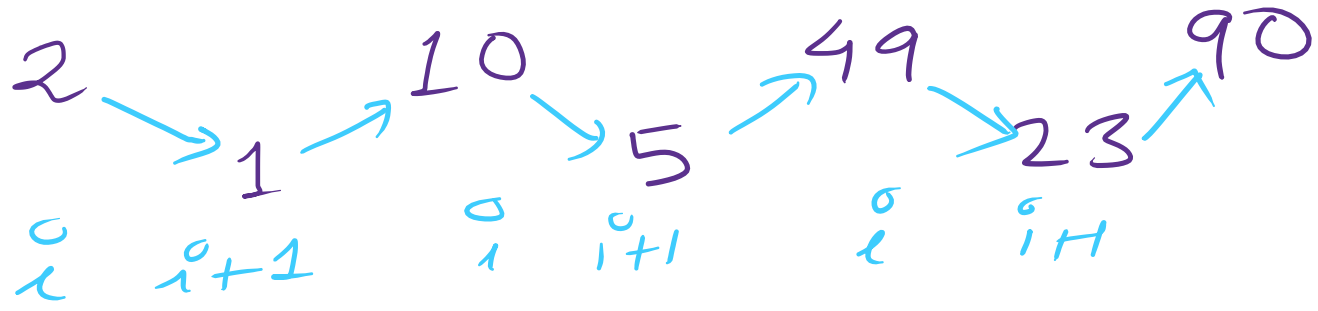
Left

$$(s, p-1)$$

partition index

Regression

Right

$$arr = 10, 90, 49, 2, 1, 5, 23$$
$$\%P = 1, 2, 5, 10, 23, 49, 90$$


Lean $\rightarrow$ AVL Trees | Balanced BST

1962 → Adelson Velsky, Landis

Report → Process of Storing Information

* Friday

✓ Trie data Structure

- * Telephone Directory → Name / Number
- * Dictionary → Words

AMAN $\rightarrow$

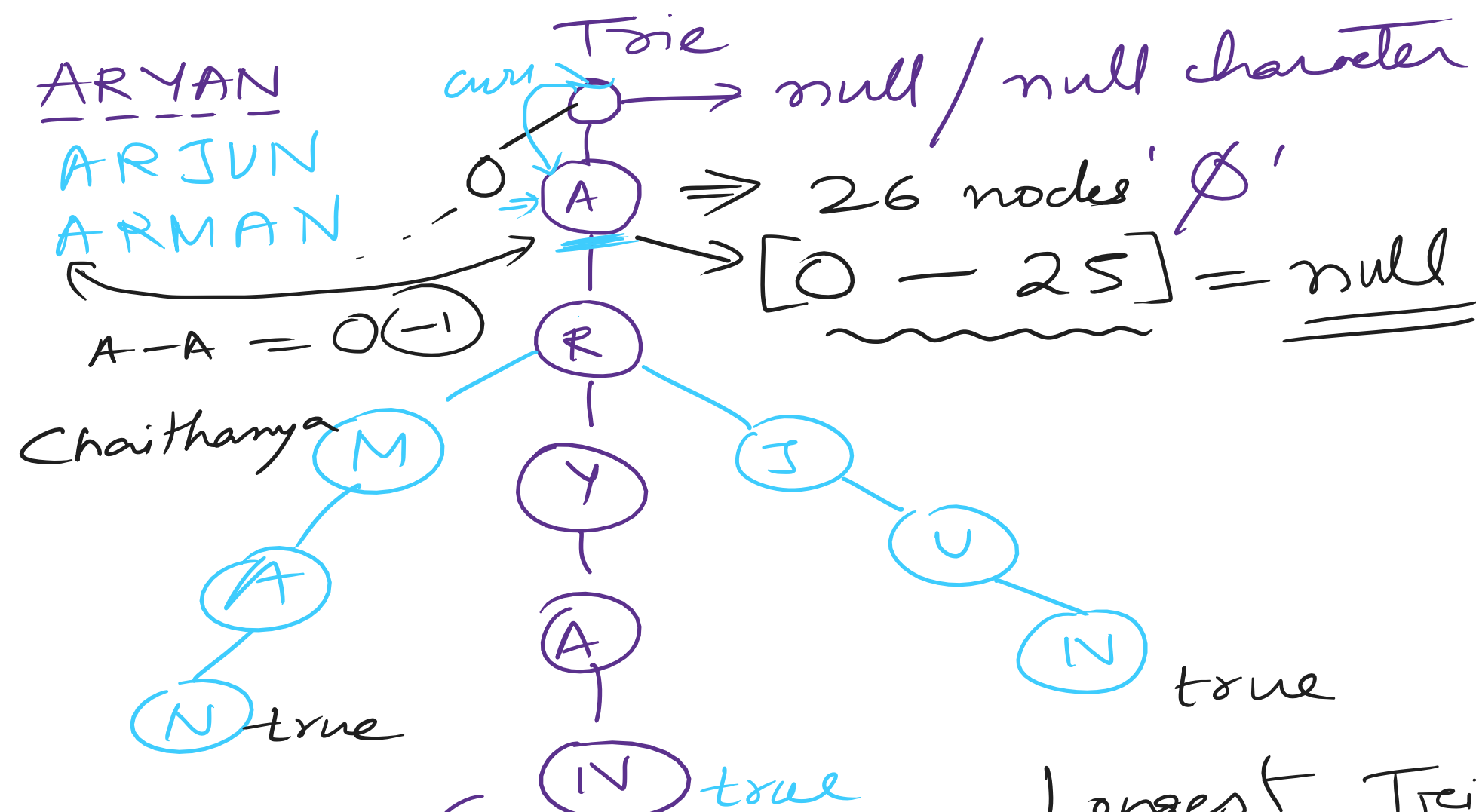
X Efficient Approach

$\langle \text{"Aman"}, 1 \rangle$
 $\langle \text{string}, \text{bool} \rangle \text{ map} \rightarrow \text{v1}$

AMAR

AMAR

$\langle \text{"amar", 1} \rangle \text{ map} \rightarrow v2$



```

class TrieNode {
    bool wordEnd = false;
    children[26] = nullptr;
}

```

Longest Trie
Common
Prefix
Suffix